Preparation, Characterization, and Simulation Prediction of MoS_2 Transistors

Kaige Yang, Qiang Gao, Xiaojing Su, Yajuan Su, Xuge Fan, and Jie Ding

***Abstract*— The advent of molybdenum disulfide (MoS_2) offers hope for transistor continuous scaling. In this study, the back-gate monolayer MoS_2 transistors at different channel lengths are prepared, and their electrical properties are tested, respectively. The back-gate MoS_2 transistor exhibits a maximum conducting current of 4.59 μA and a switching ratio of 2.3×10^5. The ratio of contact resistance to the total resistance is up to 33%, indicating that the contact between MoS_2 and the electrode is one of the main factors restricting its performance. Technology Computer-Aided Design (TCAD) simulations are also used to predict the performance of a smaller channel-length MoS_2 transistor (L_g = 12 nm). We set the MoS_2 material parameters and construct a monolayer MoS_2 nanosheet FET (NSFET) using TCAD. The simulation result of this structure shows a subthreshold swing (SS) of 68.9 mV/dec, a drain-induced barrier lowering (DIBL) value of 14.98 mV/V and a conduction current of 243.3μA/μm at the gate voltage of 0.7V. By comparing its performance with the most advanced silicon-based devices, MoS_2 exhibits excellent electrostatic properties and holds great potential for replacing silicon as a channel material.**

***Index Terms*— Bulk and nanosheet transistor, characteristic test and simulation prediction, molybdenum disulfide (MoS_2).**

I. Introduction

With the dramatic decrease in transistor size, silicon has gradually approached the scaling limit, and the influence of short channel effect (SCE) has become more and more significant [1]. Currently, the semiconductor technology node has entered sub-5 nm. According to the international roadmap for devices and systems (IRDSs), the channel thickness is limited to 6 nm [2]. At this scale, the surface roughness (SR) and the surface suspension bond of silicon seriously aggravate carrier scattering [3], [4], [5]. Also, the deterioration of subthreshold characteristics and dielectric leakage becomes increasingly prominent, resulting in higher power consumption [6], [7]. These challenges limit the performance of traditional silicon-based transistors and require a higher level of manufacturing process, and the cost increases simultaneously. In order to continue Moore's Law, one of the potential solutions is looking for alternative channel materials such as transition metal dichalcogenides (TMDCs) [4], which show excellent electrical, optical, mechanical, and thermal properties, providing a broad prospect for the next generation of electronic devices [8], [9].

MoS_2 is regarded as one of the most promising 2-D materials because of its thin atomic thickness, low dielectric constant, large bandgap, excellent carrier effective mass, and no suspended bond on the surface. MoS_2 transistors are less susceptible to SCE [10]. Studies show that at a channel length of 10 nm, monolayer MoS_2 transistors exhibit a 52% reduction in drain-induced barrier lowering (DIBL) and a 13% reduction in subthreshold swing (SS) compared to 3-nm-thick-body Si transistors [11]. In addition, MoS_2 transistor shows a big switching ratio because of its large bandgap [12], [13].

In order to demonstrate the properties of MoS_2 as the channeling material, we use experimental and simulation methods to characterize and predict the behaviors of single-layer MoS_2 transistors. In this article, the MoS_2 back-gate transistors are prepared by PMMA wet transfer, and its characteristics, such as transfer characteristics, output characteristics, threshold voltage (V_T), SS, ON/OFF current, and contact resistance, are fully characterized. In addition, we set the material parameters of $MoS_2$and built a single-layer MoS_2 nanosheet FET (NSFET) structure in Technology Computer-Aided Design (TCAD) software to capture the ideal ultimate performance of a 2-nm technology node MoS_2 NSFET. The drift–diffusion (DD) model with quantum modifications and other physical models such as the SRH model and OldSlotboom model are introduced to precisely predict the performance of MoS_2 transistors.

The arrangement of this article is as follows. In Section II, we describe the fabrication process of the MoS_2 transistor in detail. The fabricated MoS_2 transistors' electrical characteristics are fully tested and analyzed in Section III. Section IV focuses on describing the simulation methodology. Section V shows the simulation results of a 2-nm node technology MoS_2 transistor. The conclusion is drawn in Section VI.

Received 3 August 2025; revised 15 September 2025; accepted 26 September 2025. Date of publication 14 October 2025; date of current version 10 December 2025. This work was supported by Beijing Institute of Technology Research Fund Program for Young Scholars. The review of this article was arranged by Editor E. Gnani. *(Corresponding author: Jie Ding.)*

Kaige Yang is with the School of Integrated Circuits and Electronics, Beijing Institute of Technology, Beijing 100081, China.

Qiang Gao and Xuge Fan are with the School of Interdisciplinary Science, Beijing Institute of Technology, Beijing 100081, China.

Xiaojing Su and Yajuan Su are with the State Key Laboratory of Fabrication Technologies for Integrated Circuits, Institute of Microelectronics, Chinese Academy of Sciences, Beijing 100029, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China.

Jie Ding is with the School of Integrated Circuits and Electronics, Beijing Institute of Technology, Beijing 100081, China, and also with Zhengzhou Research Institute, Beijing Institute of Technology, Zhengzhou 450000, China (e-mail: dingjie2015@foxmail.com).

Digital Object Identifier 10.1109/TED.2025.3617478

1557-9646 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on February 15,2026 at 12:14:00 UTC from IEEE Xplore. Restrictions apply.

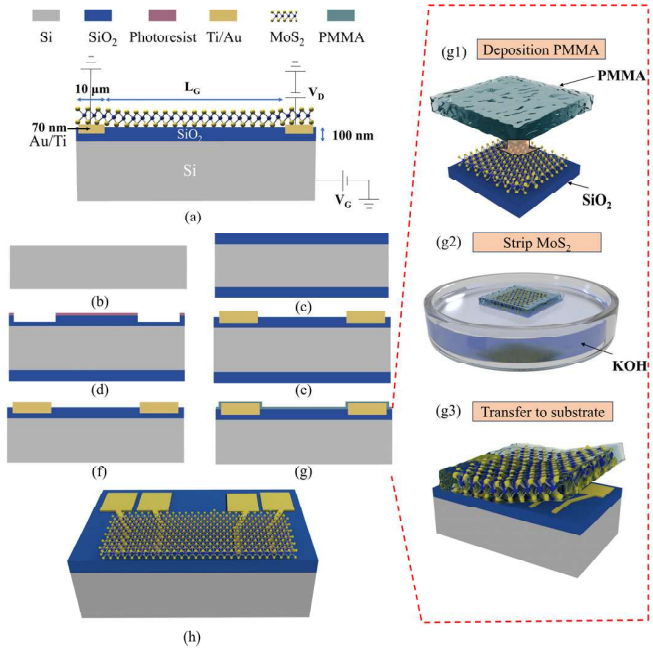

Fig. 1. Process flow of the MoS_2 back-gate transistor. (a) Cross-sectional view of the MoS_2 transistor with electrical connections. (b) Silicon wafer. (c) SiO_2 layer grown by thermal oxidation. (d) Electrode grooves prepared by reactive ion etching. (e) Electrodes prepared by thermal evaporation. (f) Bottom SiO_2 etched by fumed hydrofluoric acid. (g) MoS_2 was transferred to the target substrate. (h) 3-D schematic of the MoS_2 back-gate transistor.

II. Process Flow

The structure of the MoS_2 transistor that is fabricated in this study is shown in Fig. 1(a). It is a back-gate transistor produced on the Si substrate with a 100-nm SiO_2 dielectric. Titanium and gold serve as source and drain electrodes, with an electrode width of 10 μm. The distance between the electrodes is varied to achieve different channel lengths (15–130 μm). The MoS_2 layer forms the channel of the transistor. The Si substrate acts as a back gate, allowing for the modulation of the carrier concentration in the MoS_2 channel. The detailed process flow is divided into six steps.

1) Silicon wafers [see Fig. 1(b)] are thermally oxidized, and SiO_2 of about 100 nm is grown on both sides of the silicon [see Fig. 1(c)].
2) The photoresist (PR) is spun on the surface of SiO_2 and patterned. A 50-nm deep electrode groove is etched by reactive ion etching [see Fig. 1(d)].
3) Titanium and gold are deposited as electrodes by thermal evaporation. By controlling the distance between the electrodes on the mask, devices with various channel lengths are fabricated. The residual PR is further removed by acetone solvent dissolution and an oxygen plasma cleaning machine [see Fig. 1(e)].
4) Fumed hydrofluoric acid is used to etch away the silicon dioxide at the bottom of the substrate [see Fig. 1(f)].
5) MoS_2 (purchased from Shenzhen Six Carbon Technology) is transferred onto SiO_2 by wet transfer technology [see Fig. 1(g)].
6) Finally, by photoetching and annealing, the MoS_2 back-gate transistor is fabricated [see Fig. 1(h)].

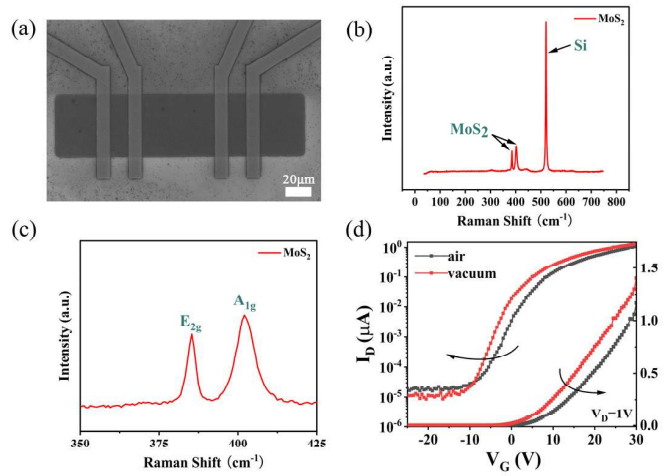

Fig. 2. Structure and characterization of MoS_2 back-gate transistor. (a) SEM image of MoS_2. (b) and (c) Raman spectra of MoS_2, the typical peaks are 385 and 402 cm^{-1}. (d) Comparison of MoS_2 transistor transfer characteristics in vacuum and air.

The wet transfer process in step 5) is as follows. First, the MoS_2 surface is spin coated with polymethyl methacrylate solution (PMMA, 950-PMMA-A4) at 500 r/min 10 s and 3000 r/min 30 s. After heating at 90 °C for 10 min, the PMMA/MoS_2/SiO_2 is placed in 1-mol/L potassium hydroxide (KOH) for 0.5 h. After three times cleaning with deionized (DI) water, the cleaned MoS_2 film is transferred to the target substrate (Au/Ti/SiO_2/Si). The whole structure is tilted at 45° until the water evaporates. The whole structure is soaked in acetone for 3 h (replace the acetone every half hour, replace six times) and in isopropanol to complete the transfer of MoS_2.

III. Experimental Results and Discussion

The scanning electron microscope (SEM) of the fabricated MoS_2 transistor at 500 magnification is shown in Fig. 2(a). Fig. 2(b) and (c) shows the Raman spectra of the MoS_2 and a magnification of the peaks, respectively. Fig. 2(b) identifies three prominent Raman peaks. The silicon Raman peak appears at 520 cm^{-1}. The remaining two Raman peaks are MoS_2, which is shown in detail in Fig. 2(c). The in-plane (E_{2g}) mode is located at 385 cm^{-1}, and the out-of-plane (A_{1g}) mode is located at 402 cm^{-1}, which indicates that the sample is MoS_2.

After characterization, the electrical properties of MoS_2 transistors are tested and analyzed. In this study, the transistors are tested with two Keithley 2450s at room temperature. The voltage sources are connected with drain–source and gate–source, as shown in Fig. 1(a). The MoS_2 transistors with a channel length of 15 μm and a width of 50 μm are tested under air and vacuum conditions, respectively. The transfer characteristic curves are shown in Fig. 2(d). Both curves exhibit prominent n-type characteristics when V_G is swept from −25 to 30 V@V_D = 1 V. Compared with the transistors in air, the ones in the vacuum environment exhibit higher I_{on}(I_{on} = 1.4 × 10^{-6} A in vacuum and 1.2 × 10^{-6} A in air), smaller I_{off} (I_{off} = 1.1 × 10^{-11} A in vacuum and 1.6 × 10^{-11} A in air) and better subthreshold characteristics (SS = 2.19 V/dec in vacuum and 2.53 V/dec in air). In addition,

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on February 15,2026 at 12:14:00 UTC from IEEE Xplore. Restrictions apply.

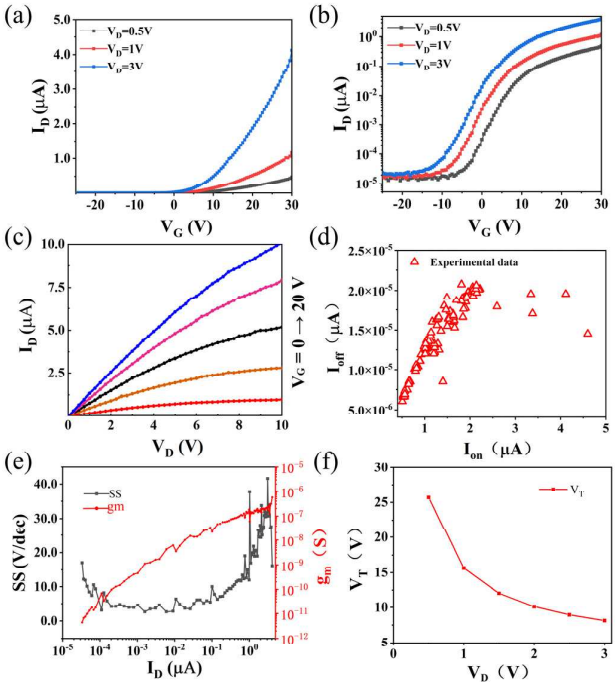

Fig. 3. Electronic characteristics of the MoS_2 transistor with channel length of 15 μm. (a) and (b) Transfer characteristics. (c) Output characteristics. (d) I_{off} as a function of I_{on}. (e) Variation of SS and g_m with I_D. (f) V_T against V_D extracted by the constant current method.

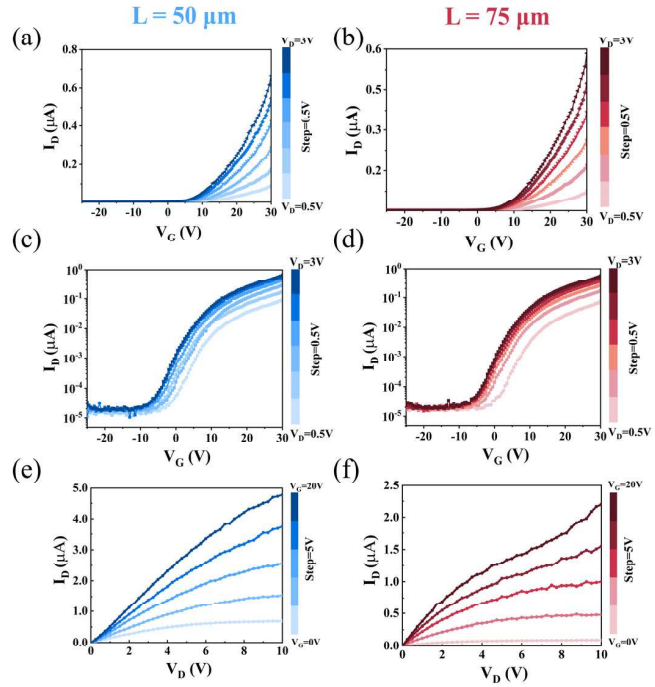

Fig. 4. Comparison of the transfer characteristics and output characteristics at different channel lengths (50 and 75 μm). (a) and (b) Comparison of the transfer characteristics at different channel lengths (50 and 75 μm) in linear. (c) and (d) Comparison of the transfer characteristics at different channel lengths (50 and 75 μm) in logarithmic. (e) and (f) Output characteristic curves at different channel lengths (50 and 75 μm).

the current of the transistors in vacuum increases rapidly when the gate voltage reaches above 0 V. It may be due to the easy adsorption of water and oxygen molecules on the surface of MoS_2 in air. It increases the probability of carrier scattering. Adsorbed impurities can form surface states that trap or scatter carriers, further affecting the transistors' I_{on}, I_{off}, and threshold characteristics.

The transistors' electrical characteristics with a channel length of 15 μm and a width of 50 μm are characterized in air. As shown in Fig. 3(a) and (b), the transfer characteristics at different V_D (0.5, 1, and 3 V) are tested. The current shows a distinct upward trend with increasing V_D. At V_D = 3 V, the current before the breakdown of the transistor could reach 4.11 μA. Fig. 3(b) shows the transfer characteristics on a logarithmic scale. Compared with V_D = 0.5 V and V_D = 1 V, I_{off} and SS of V_D = 3 V do not change significantly. I_{off} remains on the order of 10^{-11} A (1.7×10^{-11} A at V_D = 0.5 V, 1.8×10^{-11} A at V_D = 1 V and 2.4×10^{-11} A at V_D = 3 V). Similarly, the SS values show minor variations at different V_D values (2.6 V/dec at V_D = 0.5, 2.2 V/dec at V_D = 1 V, and 2.7 V/dec at V_D = 3 V). Fig. 3(c) shows the output characteristics. A total of 93 transistors are tested, and the I_{on} and I_{off} characteristics are presented in Fig. 3(d). The best-performing transistor has an I_{on} of 4.59×10^{-6} A and an I_{off} of 1.95×10^{-11} A, with an ON/OFF ratio of up to 2.3×10^5. For all the 93 tested transistors, the average values of I_{on} and I_{off} are 1.45×10^{-6} and 1.46×10^{-11} A, respectively, with the average ON/OFF ratio of 9.61×10^4. The overall switching ratio changes very little.

In order to show the sensitivity of the MoS_2 transistors to changes in V_G, the SS are calculated. The relationship between SS and I_D is shown in Fig. 3(e). SS first decreases and then increases with the increase of I_D. The minimum SS in the subthreshold region is 2.65 V/dec. Fig. 3(e) also reflects the variation of transconductance (g_m) with current, showing an overall linear trend. The maximum g_m obtained at the maximum I_D is 5.66×10^{-7} S.

Fig. 3(f) shows the threshold characteristics of the MoS_2 transistors. Since the V_T depends on V_D, V_T of transistors is calculated under a different V_D (0.5–3 V) according to the transfer characteristic curves. The results show that as V_D increases from 0.5 to 3 V, V_T exhibits a pronounced change (from 25.6 to 8.1 V).

The transfer characteristics and output characteristics of MoS_2 transistors are compared at different channel lengths (50 and 75 μm). The width of the transistor is 50 μm. Fig. 4(a) and (b) shows the transfer characteristics in linear. As channel length increases from 50 to 75 μm, the maximum saturation current decreases from 6.7×10^{-7} to 5.8×10^{-7} A. Fig. 4(c) and (d) shows the changes in the subthreshold region. At V_D = 3 V, SS varies from 2.554 V/V (channel length = 50 μm) to 2.339 V/V (channel length = 75 μm). The change of the channel length has no significant effect on SS. Fig. 4(e) and (f) shows the output characteristics of the two groups of transistors. There is a significant difference in the saturation current. At V_G = 20 V, the device with a channel length of 50 μm has a maximum saturation current of 4.78×10^{-6} A, twice of the one with the channel length of 75 μm. In addition, devices with 75-μm channel length are more likely to enter the saturation region [see Fig. 4(f)], which reduces the power consumption.

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on February 15,2026 at 12:14:00 UTC from IEEE Xplore. Restrictions apply.

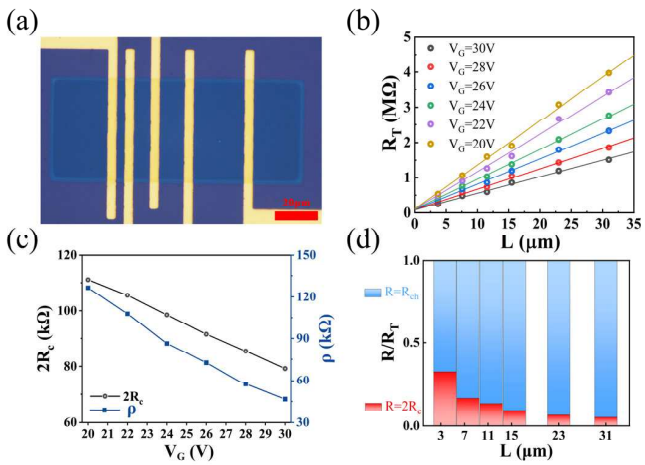

Fig. 5. (a) Optical image of the MoS_2 transistor in TLM geometry. (b) Total resistance as a function of L at various V_G. Solid lines are the linear fits. (c) Contact resistance R_c and sheet resistance ρ obtained by linear fitting at various V_G. (d) Contribution of contact resistance ($2R_c$) and channel resistance (R_{ch}) to the total resistance (R_T).

In 2-D channel devices, the contact resistance is prominent. This stems from the mismatch in Fermi levels between metals and 2-D materials. In addition, impurities and defects will be introduced to the interface between electrodes and 2-D materials during the transistor fabrication, which are also the key factors affecting the device contact resistance [14]. In the top-contact design, the metal deposition process typically involves high-energy atomic bombardment and intense local heating of the contact region, which in turn generates defects on the 2-D material and further increases the contact resistance. The bottom-contact design demonstrated here avoids the introduction of defects during metal preparation, achieving relatively low resistance. Fig. 5 shows the contact resistance characteristics of the back-contact MoS_2 transistors. The contact resistance of the MoS_2 transistors is extracted using the transfer length method (TLM) [15]. Fig. 5(a) shows the optical image of transistors with different channel lengths (L_{ch} = 3, 7, 11, 15, 23, and 31 μm). All transistors' channel width is 50 μm. As shown in Fig. 5(b), the total resistances of the transistors at different channel lengths are measured at a different V_G [(1)]. The total resistance exhibits a linear relationship with the channel length, which aligns with prior research findings [16], [17], [18]

$$R_T = \rho L_{ch} + 2R_c R_{ch} = R_T - 2R_c. \tag{1}$$

Here, R_T is the total resistance, ρ is the 2-D channel sheet resistance, L_{ch} is the channel length, R_{ch} is the channel resistance, and R_c is the contact resistance. $2R_c$ and ρ are extracted according to (1), which are shown in Fig. 5(c). $2R_c$ and ρ are linearly dependent on V_G, and the magnitude of ρ is consistent with previous reports [16]. For long channel transistors, the relationship among R_{ch}, R_T, and R_c can be expressed as (2) [15]. The stacked bar plots are used to show the contribution of $2R_c$ and R_{ch} to R_T as a function of L [see Fig. 5(d)]. The results show that R_c is very significant, and it occupies a large proportion of the total resistance in the short-channel-length transistors. When L_{ch} = 3 μm, the ratio of $2R_c$–R_T reaches 33%. However, with the increase in the channel length, the proportion of $2R_c$ gradually decreases due to the increase in channel resistance.

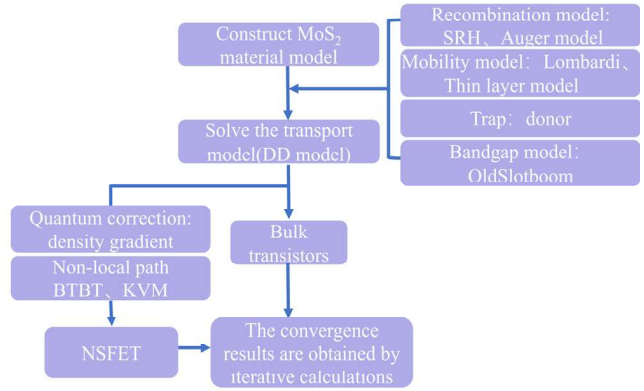

Fig. 6. Simulation flow.

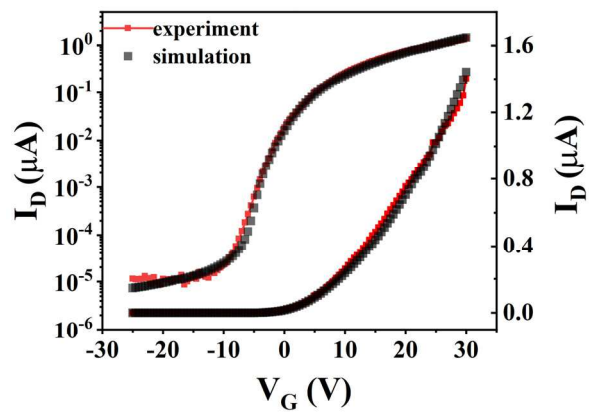

Fig. 7. Comparison of MoS_2 bulk transistor simulation results with experimental results.

TABLE I
PARAMETERS USED IN THE SIMULATION [22], [23], [24], [25], [26]

Parameter	MoS_2
Bandgap(eV)	1.8
Dielectric Constant(ε_0)	3.3
Electron Affinity(eV)	4
Effective Mass_ electron (m_0)	0.51
ConstantMobility_electron(cm^2/Vs)	102.5

IV. SIMULATION METHODOLOGY

Based on the above experimental work, we proceed to conduct the simulations of MoS_2 transistors using TCAD. The integrated approach allows to validate experimental results and to further explore the theoretical performance boundaries of the promising devices. The overall simulation process is shown in Fig. 6. The material parameters of MoS_2 are shown in Table I. It is worth mentioning that doping is introduced in the simulation to adjust the performance of single-layer MoS_2. The DD model is used to simulate the back-gate transistor. The SRH and Auger models are considered in the recombination model. Other models are also added to consider physical effects, such as the OldSlotboom model. To further match the real device, a certain concentration of defect charge is added in the channel region. The whole models are calibrated by

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on February 15,2026 at 12:14:00 UTC from IEEE Xplore. Restrictions apply.

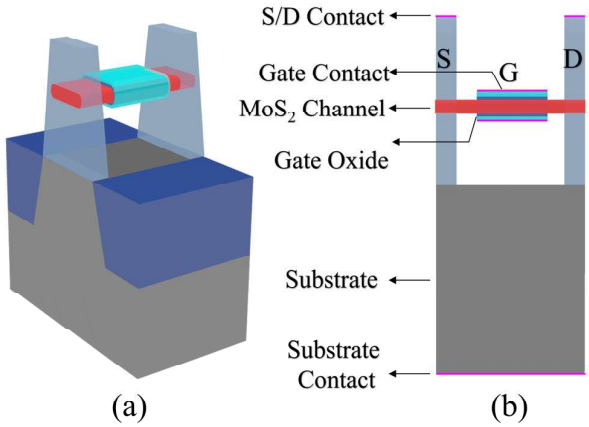

Fig. 8. (a) 3-D schematic. (b) Cross-sectional view of the MoS_2 NSFET.

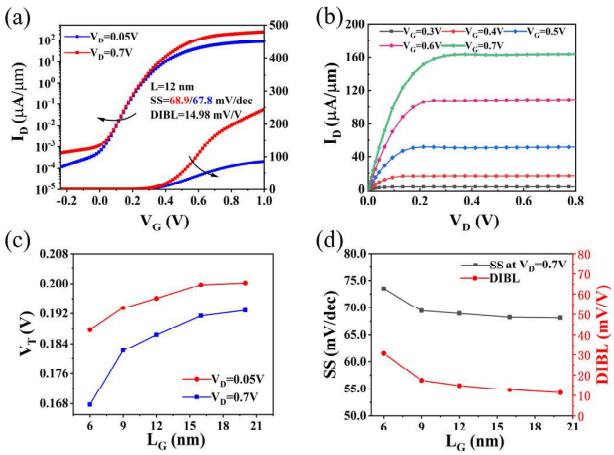

Fig. 9. Simulation results of the MoS_2 NSFET. (a) Transfer characteristics. (b) Output characteristics of the MoS_2 NSFET (channel length = 12 nm). (c) V_T as a function of L_G at a different V_D. (d) DIBL and SS as a function of L_G.

simulating the bulk transistor, which is scaled down to 0.15-μm channel length from the fabricated device. The simulation results of the bulk transistor show a good agreement with the experimental data (see Fig. 7) from Section III, validating the feasibility of the models.

Fig. 8 shows the structure of a single-layer MoS_2 NSFET with 0.6-nm $SiO_2$and 1.1-nm HfO_2 as the gate oxide. The channel thickness is 0.6 nm, and the width is 25 nm. The entire channel is n-type doped, and the concentration is 1×10^{16} cm^{-1}. The channel length in this study varies from 6 to 20 nm. The density gradient model is employed to simulate the performance of MoS_2 NSFET under the consideration of quantum effect [19], [20]. The nonlocal path band-to-band tunneling (BTBT) is added to capture the tunneling phenomena induced by reduced source–drain distance, thinned oxide layers, and Schottky contacts. In addition, the kinetic velocity model (KVM) is integrated into the simulation framework to describe the ballistic transport effects [21].

V. Simulation Prediction

Fig. 9(a) shows the transfer characteristic curve of the NSFET with a channel length of 12 nm. The device exhibits excellent DIBL (14.98 mV/V) and SS (68.9/67.8 mV/dec) close to the limit. MoS_2 NSFET shows an ultrahigh conducting current of 243.3 μA/μm at V_D = 0.7 V. The switch ratio of the NSFET has little difference with the change of V_D, also indicating a small DIBL. It is also evident that in the turn-off region, the gap between the turn-off current at high drain and low drain gradually increases as the negative V_G increases [see Fig. 9(a)]. This phenomenon may be due to the current leakage caused by the gate. Although the NSFET structure weakens the effect of the transverse electric field at significantly scaled dimensions, the gate modulates the drain band through the longitudinal electric field, thereby increasing the device's power consumption in the OFF state [27], [28]. Fig. 9(b) shows the output characteristics of the device with a channel length of 12 nm. The curves under a different V_G quickly enter the saturation region as V_D increases. At V_G = 0.7 V, the saturation current reaches 164.4 μA/μm.

Fig. 9(d) shows the comparison of the variations of SS and DIBL when channel length changes. When the channel length is decreasing from 20 to 6 nm, SS increases from 68.1 to 73.5 mV/dec, while DIBL increases from 11.4 to 30.6 mV/V. When the channel length is 6 nm, SS and DIBL mutate, and V_T changes from 0.193 V (L_g = 20 nm) to 0.168 V (Lg = 6 nm) [see Fig. 9(c)]. These phenomena mean that the SCE gradually increases when the channel length is 6 nm. The SCE of a 2-D transistor can be quantified using the natural length (λ). We thus calculate λ for the MoS_2 NSFET by the following formula [4]:

$$\lambda_{2D} = t_{\mathrm{ch}} + t_{\mathrm{ox}}\varepsilon_{\mathrm{ch}}/\varepsilon_{\mathrm{ox}}. \tag{2}$$

Here, t_{ch} is the channel thickness (0.6 nm), t_{ox} refers to the equivalent oxide thickness derived by converting the dual-layer oxides (SiO_2 = 0.6 nm with ε_{ox} = 3.9 and HfO_2 = 1.1 nm with ε_{ox} = 22) into a single-layer equivalent, and $\lambda \approx$ 1.27 nm. To avoid SCE, the NSFET's channel length should be greater than 6 $\lambda \approx$ 7.7 nm [29]. This means that when the MoS_2 NSFET's channel length is less than 7.7 nm, the characteristics of the MoS_2 NSFET will begin to deteriorate, which matches the simulation results.

According to the results predicted by the simulation, the SS and DIBL values in this work are further compared with the most advanced silicon-based transistors. The results are shown in Table II. Compared with silicon-based transistors, MoS_2 transistors have advantages in SS and DIBL when extremely scaled. This shows the scaling potential of MoS_2 as the future channel material.

TABLE II
Comparison of Different Transistors' Characteristics

Material	Structure	L_G (nm)	V_D (V)	SS (mV/dec)	DIBL (mV/V)
Si[30]	NSFET	12	0.7	73.45	66.96
Si[31]	FinFET	-	0.7	75	66
Si[32]	NSFET	14	0.7	67	23.41
Si[33]	NSFET	12 (44CPP)	0.7	75	32
Si[34]	NSFET	13	0.7	<70	-
MoS_2[30]	NSFET	12	0.7	64.18	18.67
MoS_2[26]	NSFET	16	0.7	<70	<40
MoS_2 (this work)	NSFET	12	0.7	68.9	14.98

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on February 15,2026 at 12:14:00 UTC from IEEE Xplore. Restrictions apply.

VI. Conclusion

In summary, the electrical properties of monolayer MoS_2 transistors are evaluated in detail using experiments and simulations. In the experiment, the fabrication process and device structure are well designed, and thus, the high quality transistors are fabricated. Both experimental and simulation results show a large switching ratio and conduction current, which indicates that MoS_2 transistors have excellent performance in gate control. In addition, the simulation results also reveal that MoS_2 transistors can maintain excellent gate control even under extreme scaling conditions, highlighting its great potential in transistor scaling. However, MoS_2 transistors also face some challenges, in which the impact of contact resistance on device performance is particularly significant. Both simulation and theoretical results show that when the channel length is reduced below 7.7 nm, the SCE leads to an increase in DIBL and SS, which limits its ability to scale further, providing a clear direction for future research and optimization. In general, MoS_2 exhibits excellent electrostatic characteristics, showing its potential in future electronic devices.

References

[1] A. Chaudhry and M. J. Kumar, "Controlling short-channel effects in deep-submicron SOI MOSFETs for improved reliability: A review," *IEEE Trans. Device Mater. Rel.*, vol. 4, no. 1, pp. 99–109, Mar. 2004, doi: 10.1109/TDMR.2004.824359.

[2] *International Roadmap for Devices and Systems: 2022 Update*, Inst. Elect. Electron. Eng. (IEEE), Piscataway, NJ, USA, 2022.

[3] K. Uchida, H. Watanabe, A. Kinoshita, J. Koga, T. Numata, and S. Takagi, "Experimental study on carrier transport mechanism in ultrathin-body SOI NAND p-MOSFETs with SOI thickness less than 5 nm," in *IEDM Tech. Dig.*, Dec. 2002, pp. 47–50, doi: 10.1109/IEDM.2002.1175776.

[4] Y. Liu, X. Duan, H.-J. Shin, S. Park, Y. Huang, and X. Duan, "Promises and prospects of two-dimensional transistors," *Nature*, vol. 591, no. 7848, pp. 43–53, Mar. 2021, doi: 10.1038/s41586-021-03339-z.

[5] S. Jin, M. V. Fischetti, and T.-W. Tang, "Modeling of surface-roughness scattering in ultrathin-body SOI MOSFETs," *IEEE Trans. Electron Devices*, vol. 54, no. 9, pp. 2191–2203, Sep. 2007, doi: 10.1109/TED.2007.902712.

[6] R. K. Cavin, P. Lugli, and V. V. Zhirnov, "Science and engineering beyond Moore's law," *Proc. IEEE*, vol. 100, no. Special Centennial Issue, pp. 1720–1749, May 2012, doi: 10.1109/JPROC.2012.2190155.

[7] M. Ieong, B. Doris, J. Kedzierski, K. Rim, and M. Yang, "Silicon device scaling to the sub-10-nm regime," *Science*, vol. 306, no. 5704, pp. 2057–2060, Dec. 2004, doi: 10.1126/science.1100731.

[8] C. J. Dorow et al., "Gate length scaling beyond Si: Mono-layer 2D channel FETs robust to short channel effects," in *IEDM Tech. Dig.*, San Francisco, CA, USA, Dec. 2022, pp. 7.5.1–7.5.4, doi: 10.1109/IEDM45625.2022.10019524.

[9] G. Arutchelvan et al., "Impact of device scaling on the electrical properties of MoS_2 field-effect transistors," *Sci. Rep.*, vol. 11, no. 1, Mar. 2021, Art. no. 6610, doi: 10.1038/s41598-021-85968-y.

[10] W. Jin, "High electric field transport characteristics in field-effect transistors based on monolayer/few-layer MoS_2," *IEEE Trans. Electron Devices*, vol. 70, no. 8, pp. 3992–4000, Aug. 2023, doi: 10.1109/TED.2023.3283374.

[11] L. Liu, Y. Lu, and J. Guo, "On monolayer MoS_2 field-effect transistors at the scaling limit," *IEEE Trans. Electron Devices*, vol. 60, no. 12, pp. 4133–4139, Dec. 2013, doi: 10.1109/TED.2013.2284591.

[12] S. B. Desai et al., "MoS_2 transistors with 1-nanometer gate lengths," *Science*, vol. 354, no. 6308, pp. 99–102, Oct. 2016, doi: 10.1126/science.aah4698.

[13] B. Radisavljevic, A. Radenovic, J. Brivio, V. Giacometti, and A. Kis, "Single-layer MoS_2 transistors," *Nature Nanotechnol.*, vol. 6, no. 3, pp. 147–150, Mar. 2011, doi: 10.1038/nnano.2010.279.

[14] J. Kwon et al., "200-mm-wafer-scale integration of polycrystalline molybdenum disulfide transistors," *Nature Electron.*, vol. 7, no. 5, pp. 356–364, Apr. 2024, doi: 10.1038/s41928-024-01158-4.

[15] Z. Cheng et al., "How to report and benchmark emerging field-effect transistors," *Nature Electron.*, vol. 5, no. 7, pp. 416–423, Jul. 2022, doi: 10.1038/s41928-022-00798-8.

[16] L. Xie, M. Liao, and S. Wang, "Graphene-contacted ultrashort channel monolayer MoS_2 transistors," *Adv. Mater.*, vol. 29, no. 37, 2017, Art. no. 1702522, doi: 10.1002/adma.201702522.

[17] A. Sebastian, R. Pendurthi, T. H. Choudhury, J. M. Redwing, and S. Das, "Benchmarking monolayer MoS_2 and WS_2 field-effect transistors," *Nature Commun.*, vol. 12, no. 1, Jan. 2021, Art. no. 693, doi: 10.1038/s41467-020-20732-w.

[18] S. Chen et al., "Channel and contact length scaling of two-dimensional transistors using composite metal electrodes," *Nature Electron.*, vol. 8, no. 5, pp. 394–402, May 2025, doi: 10.1038/s41928-025-013\break 82-6.

[19] H. Luo, X. Wang, and X. Miao, "2D structural variation impact on electrostatic performance of sub-5um nanosheet transistors subject to strong quantum confinement," in *Proc. 5th IEEE Electron Devices Technol. Manuf. Conf. (EDTM)*, Chengdu, China, Apr. 2021, pp. 1–3, doi: 10.1109/EDTM50988.2021.9421040.

[20] J.-P. Colinge, "Quantum-wire effects in trigate SOI MOSFETs," *Solid-State Electron.*, vol. 51, no. 9, pp. 1153–1160, Sep. 2007, doi: 10.1016/j.sse.2007.07.019.

[21] O. Penzin, L. Smith, A. Erlebach, M. Choi, and K.-H. Lee, "Kinetic velocity model to account for ballistic effects in the drift-diffusion transport approach," *IEEE Trans. Electron Devices*, vol. 64, no. 11, pp. 4599–4606, Nov. 2017, doi: 10.1109/TED.2017.2751968.

[22] B. Radisavljevic, M. B. Whitwick, and A. Kis, "Integrated circuits and logic operations based on single-layer MoS_2," *ACS Nano*, vol. 5, no. 12, pp. 9934–9938, Dec. 2011, doi: 10.1021/nn203715c.

[23] C. Kshirsagar, W. Xu, C. H. Kim, and S. J. Koester, "Design and analysis of MoS_2-based MOSFETs for ultra-low-leakage dynamic memory applications," in *Proc. 72nd Device Res. Conf.*, Santa Barbara, CA, USA, Jun. 2014, pp. 187–188, doi: 10.1109/DRC.2014.687\break 2360.

[24] P.-F. Chen, E. Chen, and Y.-R. Wu, "Design of monolayer MoS_2 nanosheet transistors for low-power applications," *IEEE Trans. Electron Devices*, vol. 69, no. 1, pp. 358–363, Jan. 2022, doi: 10.1109/TED.2021.3130840.

[25] E. Pascual, J. M. Iglesias, M. J. Martín, and R. Rengel, "Electronic transport and noise characterization in MoS_2," *Semicond. Sci. Technol.*, vol. 35, no. 5, May 2020, Art. no. 055021, doi: 10.1088/1361-6641/ab7777.

[26] Y. Shen, H. Tian, and T. Ren, "Simulation of MoS_2 stacked nanosheet field effect transistor," *J. Semicond.*, vol. 43, no. 8, Aug. 2022, Art. no. 082002, doi: 10.1088/1674-4926/43/8/082002.

[27] J. Fan, M. Li, X. Xu, Y. Yang, H. Xuan, and R. Huang, "Insight into gate-induced drain leakage in silicon nanowire transistors," *IEEE Trans. Electron Devices*, vol. 62, no. 1, pp. 213–219, Jan. 2015, doi: 10.1109/TED.2014.2371916.

[28] M. De Souza et al., "High temperature and width influence on the GIDL of nanowire and nanosheet SOI nMOSFETs," *IEEE J. Electron Devices Soc.*, vol. 11, pp. 672–680, 2023, doi: 10.1109/JEDS.2023.3264876.

[29] I. Ferain, C. A. Colinge, and J.-P. Colinge, "Multigate transistors as the future of classical metal–oxide–semiconductor field-effect transistors," *Nature*, vol. 479, no. 7373, pp. 310–316, Nov. 2011, doi: 10.1038/nature10676.

[30] A. Rawat and B. Rawat, "Multichannel two-dimensional MoS_2 nanosheet MOSFET for future technology node," *IEEE Trans. Electron Devices*, vol. 71, no. 6, pp. 3945–3951, Jun. 2024, doi: 10.1109/TED.2024.3384136.

[31] B. Sell et al., "Intel 4 CMOS technology featuring advanced FinFET transistors optimized for high density and high-performance computing," in *Proc. IEEE Symp. VLSI Technol. Circuits (VLSI Technol. Circuits)*, Jun. 2022, pp. 282–283, doi: 10.1109/VLSITechnologyandCir46769.2022.9830194.

[32] E. Mohapatra, T. P. Dash, J. Jena, S. Das, and C. K. Maiti, "Design study of gate-all-around vertically stacked nanosheet FETs for sub-7nm nodes," *Social Netw. Appl. Sci.*, vol. 3, no. 5, p. 540, May 2021, doi: 10.1007/s42452-021-04539-y.

[33] N. Loubet et al., "Stacked nanosheet gate-all-around transistor to enable scaling beyond FinFET," in *Proc. Symp. VLSI Technol.*, Kyoto, Japan, Jun. 2017, pp. T230–T231, doi: 10.23919/VLSIT.2017.7998183.

[34] J. Park et al., "First demonstration of 3-dimensional stacked FET with top/bottom source-drain isolation and stacked n/p metal gate," in *IEDM Tech. Dig.*, San Francisco, CA, USA, Dec. 2023, pp. 1–4, doi: 10.1109/iedm45741.2023.10413694.

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on February 15,2026 at 12:14:00 UTC from IEEE Xplore. Restrictions apply.